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TM Forum Specification

Intent Management Conformance Profile

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Introduction - API description

The following document is the conformance specification of the REST API for Any management.

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Resource Model Conformance

API Mandatory Resources

The following table indicates the mandatory resources for this API.

Resource Name	Comment
Intent	
IntentReport	
IntentSpecification	
ProbeIntent	

General Notes on Resource Attribute Conformance

There are three situations that could occur for an attribute:

- Mandatory attribute,
- Mandatory attribute if the optional parent attribute is present,
- Non-mandatory/Optional attribute. Those are all the other attributes not mentioned in the following subsections. Please refer to the corresponding API REST Specification for more details.
- The tables in the subsections below indicate which attributes are mandatory including mandatory when an optional parent is present.
- Where a resource is an input into an API (e.g. POST, PATCH), Mandatory means that the attribute value must be supplied by the API consumer in the input (and must not be blank or null).
- Where a resource is an output from an API (e.g. GET, POST), Mandatory means that the attribute value must be supplied by the API provider in the output (and must not be blank or null).
- For a sub-resource that is not an array, Mandatory (cardinality 1..1) means that the sub-resource must be present, while Optional (cardinality 0..1) means that the sub-resource may be absent. Mandatory and Optional on the sub-resource attributes apply to the sub-resource if present.
- For a sub-resource that is an array, Mandatory (cardinality 1..**) means that at least one sub-resource must be present in the array, while Optional (cardinality 0..**) means that the array may be absent. Mandatory and Optional on the sub-resource attributes apply to each of the sub-resources if present.

In this table are listed all mandatory attributes. A mandatory attribute MUST be retrieved in resource representation when no attribute selection is used (e.g. GET /../{id}) without any attribute selection

When an attribute is listed with an indentation (in second column) it means this attribute is mandatory if root (parent) attribute is present.

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Intent Resource Mandatory Attributes

Mandatory attribute when parent is present		Rule
@type		M
href		M (in response messages)
name		M
expression		
	@type	M
	iri	M
	expressionValue	M
attachment (if present)		
	@type	M
	attachmentType	M
	id	M
	mimeType	M
	name	M
characteristic (if present)		
	@type	M
	name	M
	value	M
characteristic.characteristicRelationship (if present)		
	@type	M
	id	M
	relationshipType	M
intentRelationship (if present)		
	@referredType	M
	@type	M
	id	M
	relationshipType	M
intentRelationship.associationSpec (if present)		
	@type	M
	id	M
	name	M
intentSpecification (if present)		See conditions for EntityRef at intentRelationship.associationSpec.
relatedParty (if present)		

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Mandatory attribute when parent is present		Rule
	@type	M
	role	M
relatedParty.partyOrPartyRole (if present)		
	@type	M
	id	M
	name	M

IntentReport Resource Mandatory Attributes

Mandatory attribute when parent is present		Rule
@type		M
creationDate		M
href		M (in response messages)
name		M
expression		
	@type	M
	iri	M
	expressionValue	M
intent.attachment (if present)		
	@type	M
	attachmentType	M
	id	M
	mimeType	M
	name	M
intent.characteristic (if present)		
	@type	M
	name	M
intent.characteristic.characteristicRelationship (if present)		
	@type	M
	id	M
	relationshipType	M
intent.expression (if present)		See conditions for IntentExpression at expression.
intent.intentRelationship (if present)		
	@referredType	M

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Mandatory attribute when parent is present		Rule
	@type	M
	id	M
	relationshipType	M
intent.intentRelationship.associationSpec (if present)		
	@type	M
	id	M
	name	M
intent.intentSpecification (if present)		See conditions for EntityRef at intent.intentRelationship.associationSpec.
intent.relatedParty (if present)		
	@type	M
	role	M
intent.relatedParty.partyOrPartyRole (if present)		
	@type	M
	id	M
	name	M

IntentSpecification Resource Mandatory Attributes

Mandatory attribute when parent is present		Rule
@type		M
href		M (in response messages)
name		M
attachment (if present)		
	@type	M
	attachmentType	M
	id	M
	mimeType	M
	name	M
constraint (if present)		
	@type	M
	id	M
	name	M
entitySpecRelationship.associationSpec		

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Mandatory attribute when parent is present		Rule
	@type	M
	id	M
	name	M
entitySpecRelationship (if present)		
	relationshipType	M
	role	M
	validFor	M
expressionSpecification (if present)		
	@type	M
intentSpecRelationship (if present)		
	@type	M
	id	M
	name	M
relatedParty (if present)		
	@type	M
	role	M
relatedParty.partyOrPartyRole (if present)		
	@type	M
	id	M
	name	M
specCharacteristic (if present)		
	@type	M
	name	M
	valueType	M
specCharacteristic.charSpecRelationship (if present)		
	@type	M
	name	M
	parentSpecificationId	M
	relationshipType	M
specCharacteristic.characteristicValueSpecification (if present)		
	@type	M
targetEntitySchema (if present)		
	@schemaLocation	M

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Mandatory attribute when parent is present		Rule
	@type	M

ProbeIntent Resource Mandatory Attributes

Mandatory attribute when parent is present		Rule
@type		M
name		M
expression		
	@type	M
	iri	M
	expressionValue	M
attachment (if present)		
	@type	M
	attachmentType	M
	id	M
	mimeType	M
	name	M
characteristic (if present)		
	@type	M
	name	M
characteristic.characteristicRelationship (if present)		
	@type	M
	id	M
	relationshipType	M
intentRelationship (if present)		
	@referredType	M
	@type	M
	id	M
	relationshipType	M
intentRelationship.associationSpec (if present)		
	@type	M
	id	M
	name	M
intentSpecification (if present)		See conditions for EntityRef at intentRelationship.associationSpec.

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Mandatory attribute when parent is present		Rule
relatedParty (if present)		
	@type	M
	role	M
relatedParty.partyOrPartyRole (if present)		
	@type	M
	id	M
	name	M

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API Operations Conformance

For every single resource the following tables includes mandatory operations.

Intent Mandatory Operations

The following table indicates which ones are mandatory for the Intent resource:

Uniform API Operation	Comment
GET	
POST	
DELETE	
PATCH	

IntentReport Mandatory Operations

The following table indicates which ones are mandatory for the IntentReport resource:

Uniform API Operation	Comment
GET	
DELETE	

IntentSpecification Mandatory Operations

The following table indicates which ones are mandatory for the IntentSpecification resource:

Uniform API Operation	Comment
GET	
POST	
DELETE	
PATCH	

ProbeIntent Mandatory Operations

There are no mandatory operations for the ProbeIntent resource.

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API GET Operation Conformance

All the GET operations in this API share the same status code pattern.

GET	Mandatory/Optional
Response Status Code 200 if successful	M
Response Status Code 404 if not found	M

Definitions for Filter

The following definitions apply to all the GET operations:

- Filtered Search: A filtered search can be applied using query parameters to obtain only the resources that meet the criteria defined by the filtering parameters included in the query request. Several elements can be applied to the filtered search. In that case logic, a logical AND is applied to combine the criteria (e.g.:?severity=<value>&status=<value>).
- Attribute selection (Limiting Response Data): In order to limit which attributes are included in the response, the GET request can include the ?fields= query parameter. Only those attributes whose names are supplied in this parameter will be returned. Attribute selection capabilities are the same for collections retrieval and individual resource queries.
- Level: The filtering and attribute selection can apply to the top level properties (attributes) and subproperties. The tables below show which attributes need to be supported in top-level or contained resources.

GET /intent?fields=…&{filtering}

List or find Intent objects.

Attribute selection is mandatory for all first level attributes except for the href attribute. Filtering on sub-resources is optional for all compliance levels

GET /intent/{id}?fields=…&{filtering}

This operation retrieves a Intent entity. Attribute selection enabled for all first level attributes.

Attribute selection is mandatory for all first level attributes except for the href attribute. Filtering on sub-resources is optional for all compliance levels

GET /intent/{intentId}/intentReport?fields=…&{filtering}

List or find IntentReport objects.

Attribute selection is mandatory for all first level attributes except for the href attribute. Filtering on sub-resources is optional for all compliance levels

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GET /intent/{intentId}/intentReport/{id}?fields=…&{filtering}

This operation retrieves a IntentReport entity. Attribute selection enabled for all first level attributes.

Attribute selection is mandatory for all first level attributes except for the href attribute. Filtering on sub-resources is optional for all compliance levels

There are no mandatory GET operations for IntentReport.

GET /intentSpecification?fields=…&{filtering}

List or find IntentSpecification objects.

Attribute selection is mandatory for all first level attributes except for the href attribute. Filtering on sub-resources is optional for all compliance levels

GET /intentSpecification/{id}?fields=…&{filtering}

This operation retrieves a IntentSpecification entity. Attribute selection enabled for all first level attributes.

Attribute selection is mandatory for all first level attributes except for the href attribute. Filtering on sub-resources is optional for all compliance levels

The **ProbeIntent** resource is an extension of the Intent resource. The rules for listing or finding ProbeIntent objects are the same as those mentioned for the Intent resource

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API POST Operation Conformance

All the POST operations in this API share the same status code pattern.

POST	Mandatory/Optional
Status Code 201 if resource created	M

POST /intent?fields=…&{filtering}

This operation creates a Intent entity.

The following table provides the list of mandatory attributes when creating an instance of the Intent resource, including any possible rule, conditions and applicable default values.

Mandatory Attribute	Rule
expression	The Expression is validated against the TMForum Intent Ontology which is a RDF based Ontology serialized in JSON-LD, Turtle, XML or YAML-LD. JsonLdExpression extension support is mandatory.
expression.@type	
expression.iri	
expression.expressionValue	
name	
@type	

There are no mandatory POST operations for IntentReport.

POST /intentSpecification?fields=…&{filtering}

This operation creates a IntentSpecification entity.

The following table provides the list of mandatory attributes when creating an instance of the IntentSpecification resource, including any possible rule, conditions and applicable default values.

Mandatory Attribute	Rule
name	
@type	

The **ProbeIntent** resource is an extension of the Intent resource. The mandatory attributes required when creating an instance of the ProbeIntent resource are the same as those listed for the Intent resource.

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API PATCH Operation Conformance

All the PATCH operations in this API share the same status code pattern.

The mandatory application context is JSON Merge.

PATCH	Mandatory/Optional
Status Code 200 if resource modified	M

PATCH /intent/{id}?fields=…&{filtering}

This operation updates partially a Intent entity.

Non Patchable Attribute	Rule
creationDate	
href	
id	
lastUpdate	
statusChangeDate	
version	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attribute	Rule
attachment	
characteristic	
context	
description	
expression	
intentRelationship	
intentSpecification	
isBundle	
lifecycleStatus	
name	
priority	
relatedParty	
validFor	

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There are no mandatory PATCH operations for **IntentReport**.

PATCH /intentSpecification/{id}?fields=…&{filtering}

This operation updates partially a IntentSpecification entity.

Non Patchable Attribute	Rule
href	
id	
lastUpdate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attribute	Rule
attachment	
constraint	
description	
entitySpecRelationship	
expressionSpecification	
intentSpecRelationship	
isBundle	
lifecycleStatus	
name	
relatedParty	
specCharacteristic	
targetEntitySchema	
validFor	
version	

The **ProbeIntent** resource is an extension of the Intent resource. The patchable and non-patchable attributes when updating an instance of the ProbeIntent resource are the same as those listed for the Intent resource.

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API DELETE Operation Conformance

All the DELETE operations in this API share the same status code pattern.

DELETE	Mandatory/Optional
Status Code 204 if resource deleted	M

DELETE /intent/{id}

This operation deletes a Intent entity.

DELETE /intent/{intentId}/intentReport/{id}

This operation deletes a IntentReport entity.

The DELETE operation is not mandatory for **IntentReport**.

DELETE /intentSpecification/{id}

This operation deletes a IntentSpecification entity.

The **ProbeIntent** resource is an extension of the Intent resource. The rules for deleting an instance of the ProbeIntent resource are the same as those listed for the Intent resource.

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Acknowledgements

Release History

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V4.0.2	September 2022	Kamal Maghsoudlou (Ericsson)	1st Release
V5.0.0	April 2024	Kamal Maghsoudlou (Ericsson)	OAS3.0 support

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